

Vision Document for "eClinician – Clinical Management System"

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1. Introduction

Many hospitals and clinics still rely on paper records or disconnected software systems to manage patient information, appointments, prescriptions, and medical records. This often results in duplicate records, scheduling conflicts, delays in accessing patient information, and inefficient communication among healthcare staff.

As healthcare facilities grow, managing clinical operations manually becomes increasingly challenging. Healthcare providers need a centralized system that enables them to efficiently register patients, schedule appointments, maintain medical records, manage prescriptions, and process laboratory requests.

The eClinician system is a web-based Clinical Management System designed to support the daily operations of healthcare facilities through a single centralized application. The system will allow authorized users to manage patients, appointments, medical records, prescriptions, and laboratory requests based on their assigned roles.

The project will be implemented as a web application using Java Spring Boot for the backend, React with TypeScript for the frontend, and PostgreSQL as the database management system.

2. Positioning

2.1 Problem Statement

The problem of	Managing patient information, appointments, medical records, prescriptions, and laboratory requests in a healthcare facility.
Affects	Hospital administrators, clinicians, receptionists, pharmacists, laboratory technicians, and patients.
The impact of which is	Duplicate patient records, scheduling conflicts, delays in accessing patient information, inefficient communication among healthcare staff, and increased administrative workload.

A successful solution would be	One web-based Clinical Management System that provides a centralized database and an easy-to-use interface for managing patients, appointments, consultations, prescriptions, and laboratory requests.
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2.2 Product Position Statement

For	Hospitals, clinics, and healthcare providers
Who	Need an efficient and secure way to manage clinical operations and patient information
The eClinician Clinical Management System	Is a web-based healthcare management application
That	Provides a centralized platform for managing patients, appointments, medical records, prescriptions, and laboratory requests
Unlike	Paper-based records or multiple independent software applications
Our product	Provides a single integrated system that improves efficiency, reduces manual work, and enables authorized healthcare staff to securely access patient information.

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Hospital Administrator

Responsible for managing users, departments, clinicians, and overall system administration.

Clinician (Doctor)

Examines patients, records diagnoses, updates medical records, requests laboratory tests, and prescribes medication.

Receptionist

Registers new patients, updates patient information, and schedules appointments.

Pharmacist

Views prescriptions issued by clinicians and records medication dispensing.

Laboratory Technician

Receives laboratory requests, records laboratory test results, and makes results available to clinicians.

Patient

Receives healthcare services through the hospital and benefits from improved clinical workflow and record management.

3.2 User Environment

The eClinician system will be used primarily within hospitals and healthcare clinics by authorized staff members during daily clinical operations.

Receptionists will use the system to register patients and schedule appointments. Clinicians will access patient medical histories, record consultations, issue prescriptions, and request laboratory tests. Pharmacists will review prescriptions before dispensing medication, while laboratory technicians will record laboratory results for clinician review. Hospital administrators will manage users and monitor overall system activities.

The application will be accessed using standard web browsers running on desktop or laptop computers connected to the hospital network. Multiple users may access the system simultaneously while working in different departments.

4. Product Overview**4.1 Product Perspective**

The eClinician Clinical Management System is an independent web-based application developed to support the daily operations of small and medium-sized healthcare facilities.

- User Authentication
- Patient Management
- Clinician Management
- Appointment Scheduling
- Medical Records Management
- Prescription Management
- Laboratory Request and Results Management

Each module shares a centralized PostgreSQL database to ensure authorized users have access to accurate and up-to-date information.

4.2 Assumptions and Dependencies

The development of the eClinician system is based on the following assumptions:

- Healthcare staff have access to computers with internet connectivity.
- Users are provided with valid login credentials.
- Patient information is entered accurately by authorized personnel.
- The system will initially be used by a single healthcare facility.
- Users possess basic computer skills.

The project uses on the following technologies:

- Java Spring Boot
- React with TypeScript
- PostgreSQL
- Maven
- Git
- IntelliJ IDEA
- Visual Studio Code

4.3 Needs and Features

Stakeholder Need	System Feature
Register new patients	Patient Registration
Maintain patient information	Patient Management
Schedule patient appointments	Appointment Scheduling
Record consultation details	Medical Records Management
Issue prescriptions	Prescription Management
Request laboratory tests	Laboratory Request Management
Record laboratory results	Laboratory Results Management
Secure access to patient information	User Authentication and Role-Based Authorization

4.4 Alternatives and Competition

Currently, many small healthcare facilities continue to use paper records for patient management. While this approach requires little technical infrastructure, it often results in misplaced records, duplicate information, and slow retrieval of patient data.

Some clinics use separate software applications for appointments, patient records, and prescriptions. Although these applications automate individual tasks, they often lack integration, requiring users to enter the same information multiple times.

The eClinician system provides a centralized solution by integrating patient management, appointments, consultations, prescriptions, and laboratory management into a single application, reducing duplication of work and improving access to information.

5. Other Product Requirements

- The application shall provide secure user authentication.
- Different users shall have access only to functions permitted by their assigned roles.
- Patient information shall be stored securely within a PostgreSQL database.
- The system shall provide an intuitive web-based user interface.
- The application shall support multiple users accessing the system simultaneously.
- The system shall provide acceptable response times during normal operation.
- The application shall be designed to allow future expansion with additional modules such as billing, insurance processing, inventory management, and reporting.
- User documentation shall be provided to describe the basic operation of the system.
- The software will be developed using Java Spring Boot, React with TypeScript, PostgreSQL, Git, and Maven.